

Section : Chemistry

Q.1 Which reagent can be employed to distinguish Benzaldehyde and Propanal ?

1. 2,4-DNP reagent
2. Fehling's Reagent
3. Tollen's Reagent
4. Baeyer's Reagent

Options

1. 1
2. 2
3. 3
4. 4

Q.2 Match List-I with List-II

List-I

List-II

Name Reaction

Reagent

(A) Rosemund Reaction

(I) CO/HCl, anhyd. AlCl_3 , CuCl

(B) Stephen Reaction

(II) CrO_2Cl_2

(C) Etard Reaction

(III) SnCl_2 and HCl

(D) Gatterman Koch Reaction

(IV) H_2 , Pd-BaSO₄

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options

1. 1
2. 2
3. 3
4. 4

Q.3 The electrolyte used in mercury cell is

1. Paste of KOH and ZnO
2. Paste of mercury oxide and C
3. NH_4Cl and ZnCl_2
4. Paste of PbO and H_2SO_4

Options

1. 1
2. 2
3. 3
4. 4

Q.4 $^{70}\text{Yb}^{2+}$ and $^{71}\text{Lu}^{3+}$ Lanthanoid ions are.....

1. Paramagnetic and diamagnetic respectively
2. Diamagnetic and diamagnetic respectively
3. Diamagnetic and paramagnetic respectively
4. Paramagnetic and paramagnetic respectively

Options

1. 1
2. 2
3. 3
4. 4

Q.5 Which of the following classes of compounds on hydrolysis yields carboxylic acids ?

- (A) RCN
- (B) RCONH_2
- (C) RCOCl
- (D) RCOOR'

Choose the \textbf{correct answer} from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.6 Arrange the following solutions in increasing order of their concentration.

- (A) 7.5 g KCl (74.5 g mol^{-1}) in 100 mL solution
(B) 3 g Urea (60 g mol^{-1}) in 1 L solution
(C) 3.6 g glucose (180 g mol^{-1}) in 100 mL solution
(D) 4 g NaOH (40 g mol^{-1}) in 1 L solution

Choose the correct answer from the options given below:

1. (A), (C), (D), (B)
2. (A), (B), (C), (D)
3. (B), (D), (C), (A).
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Which of the following statements is/are incorrect?

1. Solubility of gases is inversely proportional to the K_H value at the given pressure.
2. The value of K_H is independent of temperature.
3. Raoult law is a special case of Henry's law.
4. The solubility of a gas in a liquid varies directly with the partial pressure of gas over liquid surface.

Options 1. 1

2. 2
3. 3
4. 4

Q.8 Select the appropriate reagent for the synthesis of tert-Butyl ethyl ether

1. Tert-butyl bromide and sodium ethoxide
2. Butyl bromide and sodium tert-butoxide.
3. Propyl bromide and sodium tert-propoxide.
4. Ethyl bromide and sodium tert-butoxide.

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Match List-I with List-II

List-I

List-II

Components of binary solution

Deviation from Raoult's Law

(A) Ethanol and water

(I) Positive

(B) Chloroform and acetone

(II) No deviation

(C) n-Hexane and n-heptane

(III) Negative

(D) Carbon disulphide and acetone

(IV) Large positive

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

3. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.10 The unit of conductance(G) is

1. ohm cm
2. ohm⁻¹ cm
3. ohm⁻¹
4. ohm

Options 1. 1

2. 2
3. 3
4. 4

Q.11 What is the IUPAC name of resorcinol?

1. Benzene -1,2-diol
2. 2-methylphenol
3. 3-methylphenol
4. Benzene -1,3-diol

Options 1. 1

2. 2
3. 3
4. 4

Q.12 Which of the following statements are correct?

- (A) On dilution, conductivity of all the electrolytes decreases.
(B) On dilution, the number of ions per unit volume decreases.
(C) Molar conductivity increases with dilution.
(D) Limiting molar conductivity is the conductivity at a very high concentration.

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (A), (B) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.13 Match List-I with List-II

Aqueous Solutions

Depression in freezing point (ΔT_f) expression

- | | |
|--|----------------|
| (A) 0.2 mol kg ⁻¹ Al ₂ (SO ₄) ₃ | (I) $2K_f$ |
| (B) 0.3 mol kg ⁻¹ Na ₂ SO ₄ | (II) K_f |
| (C) 1 mol kg ⁻¹ KCl | (III) $0.9K_f$ |
| (D) 0.1 mol kg ⁻¹ K ₃ [Fe(CN) ₆] | (IV) $0.4K_f$ |

Choose the correct option:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.14 A first order reaction is 50% completed in 1.26×10^{14} s.
How much time would it take for 100% completion?

1. 3.78×10^{14} s
2. 2.52×10^{14} s
3. infinite
4. 2.52×10^{28} s

Options 1. 1

2. 2
3. 3
4. 4

Q.15 Which transition metal group is known as coinage metals?

1. group of Co
2. group of Zn
3. group of Fe
4. group of Cu

Options 1. 1

2. 2
3. 3
4. 4

Q.16 Arrange the following in increasing order of their basic strength in aqueous solution:

- (A) CH_3NH_2
- (B) $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$
- (C) $\text{C}_6\text{H}_5\text{NH}_2$
- (D) $(\text{CH}_3)_3\text{N}$

Choose the \textbf{correct} answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.17 The osmotic pressure of the equimolar solutions of the following at the same temperature will increase in the order:

- (A) NaCl
- (B) K_2SO_4
- (C) $\text{K}_3[\text{Fe}(\text{CN})_6]$
- (D) NH_2CONH_2

Choose the correct answer from the options given below:

1. (A), (C), (B), (D)
2. (D), (B), (C), (A)
3. (C), (B), (A), (D)
4. (D), (A), (B), (C)

Options 1. 1

2. 2
3. 3
4. 4

Q.18 Nucleotide is obtained by linking nucleoside to phosphoric acid at which position of sugar ?

1. 3'
2. 5'
3. 2'
4. 1'

Options 1. 1

2. 2
3. 3
4. 4

Q.19

The order of reaction with rate law expression $\frac{d[R]}{dt} = -k$ is:

1. 0
2. 1
3. 2
4. 3

Options

1. 1
2. 2
3. 3
4. 4

Q.20 Which position of phenol is involved in coupling reaction with benzene diazonium chloride ?

1. ortho
2. meta
3. para
4. both ortho and para

Options

1. 1
2. 2
3. 3
4. 4

Q.21 Activation energy of a chemical reaction can be determined by

1. determining the rate constant at standard temperature
2. determining the rate constant at two temperatures
3. determining the probability of collision
4. using catalyst

Options

1. 1
2. 2
3. 3
4. 4

Q.22 Ortho and Para nitrophenols can be separated by steam distillation. This is because

1. o-nitrophenol undergoes intramolecular hydrogen bonding
2. p-nitrophenol undergoes intramolecular hydrogen bonding
3. o-nitrophenol undergoes intermolecular hydrogen bonding
4. o-nitrophenol is water soluble

Options

1. 1
2. 2
3. 3
4. 4

Q.23 Match List-I with List-II

List-I

List-II

Structure of ether

Common name

(A) $C_6H_5OCH_2CH_3$

(I) Methyl n-propyl ether

(B) $C_6H_5O(CH_2)_6CH_3$

(II) Methyl phenyl ether

(C) $C_6H_5OCH_3$

(III) n-Heptyl phenyl ether

(D) $CH_3OCH_2CH_2CH_3$

(iv) Ethyl phenyl ether

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options

1. 1

2. 2
3. 3
4. 4

Q.24 A dipositive transition metal ion with atomic number 25 in aqueous solution; what is the magnetic moment of it?

1. 2.84 BM
2. 5.92 BM
3. 0
4. 3.87 BM

Options

1. 1
2. 2
3. 3
4. 4

Q.25 The ammonia derivatives of carbonyl compounds obtained as colorless precipitate are

- (A) Oxime
- (B) Hydrazone
- (C) Phenyl hydrazone
- (D) Semicarbazone

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.26 The α and β forms of glucose are known as

1. Epimers
2. Anomers
3. Enantiomers
4. Diastereomers

Options

1. 1
2. 2
3. 3
4. 4

Q.27 The correct expression and unit of molar conductivity (Λ_m) are:

1. $\Lambda_m = \kappa c$; $\text{ohm}^{-1} \text{ mol L}^{-1}$
2. $\Lambda_m = \kappa V$; $\text{ohm}^{-1} \text{ cm}^2$
3. $\Lambda_m = \kappa V$; $\text{ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$
4. $\Lambda_m = \kappa c$; ohm mol^{-1}

Options

1. 1
2. 2
3. 3
4. 4

Q.28 Hoffmann bromamide degradation method is used for preparation of

1. Primary amines
2. Secondary amines
3. Tertiary amines
4. Both Primary and Secondary amines

Options

1. 1
2. 2
3. 3
4. 4

Q.29 Allyl phenyl ether can be obtained by heating.

1. $\text{CH}_2=\text{CHBr}$ and $\text{C}_6\text{H}_5\text{ONa}$

2. $\text{CH}_2=\text{CHCH}_2\text{Br}$ and $\text{C}_6\text{H}_5\text{ONa}$

3. $\text{CH}_2=\text{CHBr}$ and $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2\text{ONa}$

4. $\text{CH}_2=\text{C}=\text{CHBr}$ and $\text{C}_6\text{H}_5\text{ONa}$

Options 1. 1

2. 2

3. 3

4. 4

Q.30 Considering the stability of Cu(II) and Cu(I) which of the following statements is true?

1. Cu(II) is more stable

2. stability of both Cu(II) and Cu(I) depends upon the nature of copper salt

3. Cu(II) is less stable

4. both Cu(II) and Cu(I) are equally stable

Options 1. 1

2. 2

3. 3

4. 4

Q.31 $2\text{A} + \text{B} \rightarrow \text{A}_2\text{B}$

For the above reaction, the rate law equation is, $\text{rate} = k[\text{A}][\text{B}]^2$ with $k = 2.0 \times 10^{-6} \text{ mol}^{-2} \text{ L}^2 \text{ s}^{-1}$. If the initial

concentrations of A and B were 0.1 mol L^{-1} and 0.2 mol L^{-1} , respectively, the rate of reaction after [A] is reduced to 0.06

mol L^{-1} will be

1. $3.07 \times 10^{-9} \text{ mol L}^{-1} \text{ s}^{-1}$

2. $3.89 \times 10^{-9} \text{ mol L}^{-1} \text{ s}^{-1}$

3. $4.80 \times 10^{-9} \text{ mol L}^{-1} \text{ s}^{-1}$

4. $8.0 \times 10^{-9} \text{ mol L}^{-1} \text{ s}^{-1}$

Options 1. 1

2. 2

3. 3

4. 4

Q.32 Which of the following sugars will not reduce Fehling's reagent?

1. Maltose

2. Sucrose

3. Lactose

4. Fructose

Options 1. 1

2. 2

3. 3

4. 4

Q.33 The compound which will not respond to Fehling's test is

1. Glucose

2. Ethanal

3. Benzaldehyde

4. Formaldehyde

Options 1. 1

2. 2

3. 3

Q.34 Match List-I with List-II

List-I

List-II

(Vitamins)

(Diseases)

(A) Vitamin K (I) Pernicious anaemia

(B) Vitamin A (II) Osteomalacia

(C) Vitamin B12 (III) Increase blood clotting time

(D) Vitamin D (IV) Night blindness

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)

3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options

1. 1

2. 2

3. 3

4. 4

Q.35 The amine which will not undergo benzoylation in alkaline medium is

1. Methanamine

2. N-Ethylethanamine

3. Benzenamine

4. N.N-Dimethylaniline

Options

1. 1

2. 2

3. 3

4. 4

Q.36 The correct sequence of conductivity of the following materials at 300 K in the increasing order is

(A) Copper

(B) Graphite

(C) Germanium

(D) Silicon

Choose the correct answer from the options given below:

1. (D), (B), (C), (A)

2. (A), (D), (C), (B)

3. (A), (B), (C), (D)

4. (D), (C), (B), (A)

Options

1. 1

2. 2

3. 3

4. 4

Q.37 Select the Lanthanoids which show +2 oxidation state?

(A) Yb

(B) Ce

(C) Eu

(D) Ho

Choose the correct answer from the options given below:

1. (A), and (C) only

2. (A), (B) and (C) only

3. (A), (B), (C) and (D)

4. (D) only

Options

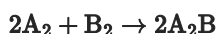
1. 1

2. 2

3. 3

4. 4

Q.38 For a reaction



The reactant A will be consumed at:

1. half the rate that B is consumed.

2. same rate at which B is consumed.

3. double the rate at which B is consumed.

4. double the rate at which A_2B is formed.

Options

1. 1
2. 2
3. 3
4. 4

Q.39 The correct statement(s) regarding the standard electrode potential, E° are
(A) The standard electrode potential changes with the change in the concentration of electrolytes
(B) The standard electrode potential is always zero at 273 K
(C) The concentration of each ion in the solution is unity.
(D) All the gases involved are at 1 bar

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (C) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.40 Arrange the following compounds in increasing order of their basicity

- (A) Aniline
- (B) p-Nitroaniline
- (C) p-Toluidine
- (D) p-Methoxyaniline

Choose the correct answer from the options given below:

1. (B), (A), (D), (C)
2. (A), (B), (C), (D)
3. (B), (A), (C), (D)
4. (A), (B), (D), (C)

Options

1. 1
2. 2
3. 3
4. 4

Q.41 Read the passage carefully and answer the Questions

Werner in 1898, propounded his theory of coordination compounds. In coordination compounds metals show two types of linkages (valences)-primary and secondary. The primary valences are normally ionisable and are satisfied by negative ions. The secondary valences are non-ionisable. These are satisfied by neutral molecules or negative ions. The secondary valence is equal to the coordination number and is fixed for a metal. The ions or group bound by the secondary linkages to the metal have characteristic spatial arrangements corresponding to the different coordination numbers. Both double salts as well as complexes are formed by the combination of two or more stable compounds in stoichiometric ratio. However, they differ in the fact that double salts such as carnallite, Mohr's salt, potash alum etc. dissociate into simple ions completely when dissolved in water. But complex ions do not dissociate completely into simple ions.

Identify the correct formula of Mohr's salt and potash alum, respectively.

1. $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$ and $KAl(SO_4)_2 \cdot 12H_2O$
2. $Fe(SO_4)_2 \cdot (NH_4)_2SO_4 \cdot 6H_2O$ and $KAl(SO_4)_2 \cdot 12H_2O$
3. $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 8H_2O$ and $KAl(SO_4)_2 \cdot 6H_2O$
4. $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 12H_2O$ and $KAl(SO_4)_3 \cdot 6H_2O$

Options

1. 1
2. 2
3. 3
4. 4

Q.42 Read the passage carefully and answer the Questions

Werner in 1898, propounded his theory of coordination compounds. In coordination compounds metals show two types of linkages (valences)-primary and secondary. The primary valences are normally ionisable and are satisfied by negative ions. The secondary valences are non-ionisable. These are satisfied by neutral molecules or negative ions. The secondary valence is equal to the coordination number and is fixed for a metal. The ions or group bound by the secondary linkages to the metal have characteristic spatial arrangements corresponding to the different coordination numbers. Both double salts as well as complexes are formed by the combination of two or more stable compounds in stoichiometric ratio. However, they differ in the fact that double salts such as carnallite, Mohr's salt, potash alum etc. dissociate into simple ions completely when dissolved in water. But complex ions do not dissociate completely into simple ions.

When 0.1 mol of $\text{CoCl}_3(\text{NH}_3)_5$ is treated with excess of AgNO_3 , 0.2 mol of AgCl are obtained. The conductivity of the

solution will correspond to

1. 1:3 electrolyte
2. 1:2 electrolyte
3. 1:1 electrolyte
4. 2:1 electrolyte

Options

1. 1
2. 2
3. 3
4. 4

Q.43 Read the passage carefully and answer the Questions

Werner in 1898, propounded his theory of coordination compounds. In coordination compounds metals show two types of linkages (valences)-primary and secondary. The primary valences are normally ionisable and are satisfied by negative ions. The secondary valences are non-ionisable. These are satisfied by neutral molecules or negative ions. The secondary valence is equal to the coordination number and is fixed for a metal. The ions or group bound by the secondary linkages to the metal have characteristic spatial arrangements corresponding to the different coordination numbers. Both double salts as well as complexes are formed by the combination of two or more stable compounds in stoichiometric ratio. However, they differ in the fact that double salts such as carnallite, Mohr's salt, potash alum etc. dissociate into simple ions completely when dissolved in water. But complex ions do not dissociate completely into simple ions.

Which of the following statement is correct as per Werner theory?

1. Secondary valences are ionisable.
2. the number of groups bound directly to the metal ion corresponding to the coordination number are Secondary valences
3. primary valences are non-ionisable
4. primary valences correspond to the coordination number of the metal

Options

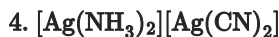
1. 1
2. 2
3. 3
4. 4

Q.44 Read the passage carefully and answer the Questions

Werner in 1898, propounded his theory of coordination compounds. In coordination compounds metals show two types of linkages (valences)-primary and secondary. The primary valences are normally ionisable and are satisfied by negative ions. The secondary valences are non-ionisable. These are satisfied by neutral molecules or negative ions. The secondary valence is equal to the coordination number and is fixed for a metal. The ions or group bound by the secondary linkages to the metal have characteristic spatial arrangements corresponding to the different coordination numbers. Both double salts as well as complexes are formed by the combination of two or more stable compounds in stoichiometric ratio. However, they differ in the fact that double salts such as carnallite, Mohr's salt, potash alum etc. dissociate into simple ions completely when dissolved in water. But complex ions do not dissociate completely into simple ions.

Which of the following is a double salt?

1. $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
2. $[\text{CrCl}_2(\text{H}_2\text{O})_4]\text{Cl} \cdot 2\text{H}_2\text{O}$
3. $[\text{FeCl}(\text{H}_2\text{O})_5]\text{Cl}_2 \cdot \text{H}_2\text{O}$



Options 1. 1

2. 2
3. 3
4. 4

Q.45 Read the passage carefully and answer the Questions

Werner in 1898, propounded his theory of coordination compounds. In coordination compounds metals show two types of linkages (valences)-primary and secondary. The primary valences are normally ionisable and are satisfied by negative ions. The secondary valences are non-ionisable. These are satisfied by neutral molecules or negative ions. The secondary valence is equal to the coordination number and is fixed for a metal. The ions or group bound by the secondary linkages to the metal have characteristic spatial arrangements corresponding to the different coordination numbers. Both double salts as well as complexes are formed by the combination of two or more stable compounds in stoichiometric ratio. However, they differ in the fact that double salts such as carnallite, Mohr's salt, potash alum etc. dissociate into simple ions completely when dissolved in water. But complex ions do not dissociate completely into simple ions.

When 1 mol $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ is treated with excess of AgNO_3 , 3 mol of AgCl are obtained. The formula of the complex is:

1. $[\text{CrCl}_3(\text{H}_2\text{O})_3] \cdot 3\text{H}_2\text{O}$
2. $[\text{CrCl}_2(\text{H}_2\text{O})_4]\text{Cl} \cdot 2\text{H}_2\text{O}$
3. $[\text{CrCl}(\text{H}_2\text{O})_5]\text{Cl}_2 \cdot \text{H}_2\text{O}$
4. $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Read the passage carefully and answer the Questions

Haloalkanes undergo nucleophilic substitution reactions owing to the polarity of C-X bond. The nucleophile reacts with the haloalkane on the carbon possessing a partial positive charge holding the halogen atom. The halogen atom X is replaced by a nucleophile. Depending on the kinetics and mode of bond breaking, the mechanism can be either SN_1 or SN_2 reaction. The rate of SN_1 reaction is governed by the stability of carbocation and in SN_2 reaction, the rate of reaction is governed by steric factor. Chirality is the main factor in both SN_1 and SN_2 . In SN_1 reaction, the chirality of alkyl halide is accompanied by racemization of the product, while in SN_2 reaction, the product is characterized by inversion of configuration. The structure of alkyl halide and the nature of the solvent also governs the mechanism of the substitution.

What is chiral carbon ?

1. Forms Superimposable mirror images
2. Attached to four different atoms or groups
3. Optically inactive
4. Symmetrical carbon

Options 1. 1

2. 2
3. 3
4. 4

Q.47 Read the passage carefully and answer the Questions

Haloalkanes undergo nucleophilic substitution reactions owing to the polarity of C-X bond. The nucleophile reacts with the haloalkane on the carbon possessing a partial positive charge holding the halogen atom. The halogen atom X is replaced by a nucleophile. Depending on the kinetics and mode of bond breaking, the mechanism can be either SN_1 or SN_2 reaction. The rate of SN_1 reaction is governed by the stability of carbocation and in SN_2 reaction, the rate of reaction is governed by steric factor. Chirality is the main factor in both SN_1 and SN_2 . In SN_1 reaction, the chirality of alkyl halide is accompanied by racemization of the product, while in SN_2 reaction, the product is characterized by inversion of configuration. The structure of alkyl halide and the nature of the solvent also governs the mechanism of the substitution.

What is meant by the term racemic mixture?

1. Can be separated by physical methods
2. An equimolar mixture of a pair of enantiomers.
3. Possess different physical properties
4. It is optically active

Options 1. 1

2. 2

3. 3

4. 4

Q.48 Read the passage carefully and answer the Questions

Haloalkanes undergo nucleophilic substitution reactions owing to the polarity of C-X bond. The nucleophile reacts with the haloalkane on the carbon possessing a partial positive charge holding the halogen atom. The halogen atom X is replaced by a nucleophile. Depending on the kinetics and mode of bond breaking, the mechanism can be either SN1 or SN2 reaction. The rate of SN1 reaction is governed by the stability of carbocation and in SN2 reaction, the rate of reaction is governed by steric factor. Chirality is the main factor in both SN1 and SN2. In SN1 reaction, the chirality of alkyl halide is accompanied by racemization of the product, while in SN2 reaction, the product is characterized by inversion of configuration. The structure of alkyl halide and the nature of the solvent also governs the mechanism of the substitution.

Which of the following bromo isomer of butane will react at a faster rate towards SN2 reaction?

1. t-butyl bromide
2. Sec-butyl bromide
3. Isobutyl bromide
4. n-butyl bromide

Options 1. 1

2. 2

3. 3

4. 4

Q.49 Read the passage carefully and answer the Questions

Haloalkanes undergo nucleophilic substitution reactions owing to the polarity of C-X bond. The nucleophile reacts with the haloalkane on the carbon possessing a partial positive charge holding the halogen atom. The halogen atom X is replaced by a nucleophile. Depending on the kinetics and mode of bond breaking, the mechanism can be either SN1 or SN2 reaction. The rate of SN1 reaction is governed by the stability of carbocation and in SN2 reaction, the rate of reaction is governed by steric factor. Chirality is the main factor in both SN1 and SN2. In SN1 reaction, the chirality of alkyl halide is accompanied by racemization of the product, while in SN2 reaction, the product is characterized by inversion of configuration. The structure of alkyl halide and the nature of the solvent also governs the mechanism of the substitution.

Which of the following compound will undergo S_N1 reaction readily?

1. $\text{CH}_3 - \text{X}$
2. $(\text{CH}_3)_2\text{CH} - \text{X}$
3. $\text{CH}_3\text{CH}_2 - \text{X}$
4. $\text{C}_6\text{H}_5\text{CH}_2 - \text{X}$

Options 1. 1

2. 2

3. 3

4. 4

Q.50 Read the passage carefully and answer the Questions

Haloalkanes undergo nucleophilic substitution reactions owing to the polarity of C-X bond. The nucleophile reacts with the haloalkane on the carbon possessing a partial positive charge holding the halogen atom. The halogen atom X is replaced by a nucleophile. Depending on the kinetics and mode of bond breaking, the mechanism can be either SN1 or SN2 reaction. The rate of SN1 reaction is governed by the stability of carbocation and in SN2 reaction, the rate of reaction is governed by steric factor. Chirality is the main factor in both SN1 and SN2. In SN1 reaction, the chirality of alkyl halide is accompanied by racemization of the product, while in SN2 reaction, the product is characterized by inversion of configuration. The structure of alkyl halide and the nature of the solvent also governs the mechanism of the substitution.

Hydrolysis of 2-Chlorobutane will result in formation of

1. (+) 2-Butanol
2. (-) 2-butanol
3. ($\pm$) 2-Butanol
4. 2-Butanol

Options 1. 1

2. 2

3. 3

4. 4

